negative numbers are good. We count the number of patients who have each value of blood pressure change, so the peak at –3 for the control group means that most of those patients had a blood pressure drop of 3 mmHg. You can see that there are two subgroups of patients in the treatment group: one group had a significant reduction in blood pressure, and another group had minimal to no effect. We call this a bimodal distribution, where mode is another word for “peak” in the distribution.

Figure 7.3. Here we compare simulated Drug A to Drug X to see which lowers blood pressure the most. Drug A has a lower average (expected) value of –15.5 mmHg and a lower standard deviation, but Drug X is bimodal with one mode centered at –22.5 mmHg. Notice that for Drug X virtually no patients had a blood pressure change that fell near the average value.

Figure 7.4. Top: An ordinary Q function takes a state-action pair and computes the associated Q value. Middle: A distributional Q function takes a state-action pair and computes a probability distribution over all possible Q values. Probabilities are bounded in the interval [0,1], so it returns a vector with all elements in [0,1] and their sum is 1. Bottom: An example Q value distribution produced by the distributional Q function for three different actions for some state. Action A is likely to lead to an average reward of –5, whereas action B is likely to lead to an average reward of +4.

Figure 7.5. Bayesian inference is the process of starting with a prior distribution, receiving some new information, and using that to update the prior into a new, more informed distribution called the posterior distribution.

Figure 7.6. Left: A discrete distribution is like a numpy array of probabilities associated with another numpy array of outcome values. There is a finite set of probabilities and outcomes. Right: A continuous distribution represents an infinite number of possible outcomes, and the y axis is the probability density (which is the probability that the outcome takes on a value within a small interval).

Figure 7.7. Screenshot from the Atari game Freeway. The objective is to move the chicken across the freeway, avoiding oncoming traffic.